

IOT Based Smart Walking Cane For Obstacle Detection And Safety Enhancement

Cohort - 5

Integration of Accelerometer

What is an Accelerometer?

- An accelerometer is a sensor that measures acceleration, which means changes in speed or movement in different directions.
- It is commonly used to detect motion, tilt, vibration, and orientation in electronic devices.

How it helps our project?

- Detects fall to the user
- Identifies abnormal tilt or cane position
- Monitors movement patterns
- Enhance safety through real time alerts

Working Principle :

- The accelerometer continuously measures movement and tilt of the walking cane in three directions (X, Y, Z).
- The microcontroller reads this data and checks if it crosses a set threshold for sudden movement or fall.
- If an abnormal condition is detected, the system activates an alert like a buzzer or sends a notification for safety.

Integration of GSM and Accelerometer

